

Data Quality Report

1. Introduction

This project intends to assess information on residential properties purchased in Ireland since 2010. We use data collected by the Residential Property Price Register (RPPR) and look into a subset of the data. This Data Quality Report will go through each feature of the sub-dataset and provide a summary of key issues within the data and potential strategies to deal with them.

2. Findings and Strategies

In this part of the report, we will analyze each feature, check their integrity, and summarize with main problems of each feature and potential solution to them. Table 1 shows an overview of our assessment of each feature, which will be elaborated in this part.

2.1 DateofSale(dd/mm/yyyy)

Initial input of *DateofSale(dd/mm/yyyy)* is viewed as an object variable because all of the input are strings of dates. From our dataset, residential property transactions span from 2016 to 2024. That brings to the key issue of *DateofSale(dd/mm/yyyy)*: raw record of this feature is too precise to provide valuable insights of our research so far. Cardinality of this feature is 2482, which is relatively high. That means we have random sales dates between 2016 to 2024, making it hard to converge upon specific trends on each single day and reducing data interpretability. Keeping the feature this way may negatively affect later analysis, specifically making it discrete or less trustworthy. To address this issue, we propose to extract month and year out of the original input. Aggregating transactions to a monthly level is more appropriate for analyzing property market trends and compare across periods.

2.2 Address

Feature *Address* is treated as an object variable because all of the input are strings of address. Similarly to *DateofSale(dd/mm/yyyy)*, *Address* also reports a high cardinality of 53357, nearly 99% of the dataset. That means, if we solely assess the dataset on *Address*, we will analyze single and separate properties. So, a similar problem is found here: *Address* is so scattered that it cannot provide aggregated trends. Since location is an important deciding factor regarding residential properties, instead of dropping the feature, we need to aggregate *Address* to identify regional trends. To solve this issue under Irish context, it can be helpful to convert original input into Eircode. Starting from Eircode, we can identify regional housing market trends.

2.3 County

Feature *County* is treated as a categorical variable because inputs are finite values, i.e. Irish counties in our context. *County* reports no significant issues shown here. There are no missing entries. Cardinality is 26, matching with administrative division of Ireland. As location-related features are important when considering housing market, we propose to keep this feature for further investigation. To have a deeper analysis, Figure 1 plots out this feature. It is clear that most transactions are concentrated in Dublin, while Cork and Kildare respectively, have the second and third highest number of house transactions. This figure also shows the necessity to look into Dublin and compare trends across different districts within Dublin.

2.4 Eircode

Feature *Eircode* is viewed as an object variable as inputs are strings of postcode. There are two issues that are worth discussing. Firstly, the missing rate of this entry is reported as 68.86%, probably RPPR treated it as an optional input. This may negatively impact later research because Eircode is a common way to extract regional information, but only assessing the 30% valid input does not lead to a robust conclusion. In this way, we propose to combine with the *Address* feature, try to fill more missing Eircode from exact addresses.

Secondly, cardinality of *Eircode* feature records as 16,761, which is relatively high. That means we will assess 16,761 different buildings, making it harder to identify aggregated trends. To address this, it is helpful to extract district information from *Eircode*, and to see if this can lead to regional trends for a certain area.

2.5 Price

Feature *Price* is considered as a continuous feature as it refers to the monetary value of residential property transactions. According to Figure 2, it is reported that property prices range from €5,250 to €221,942,731, with a mean value of € 330,160.11.

From the Histogram in Figure 2, the distribution of prices appears to be right-skewed. Most property transactions are concentrated within a relatively moderate price range, while a small number of properties have extremely high prices, creating a long right tail. This leads to a common pattern in residential property market because luxury properties can significantly increase the upper bound of prices while majority prices stay in a certain range.

From the Boxplot in Figure 2, we identify outliers and extreme values for *Price* feature. A large number of outliers is observed on the upper side of the distribution. The maximum value of €221,942,731 appears to be an extreme value. This represents either a data entry error, a bulk transaction involving multiple properties, or a special case that does not reflect typical residential transactions. Such extreme value may distort statistical analysis, especially when calculating averages or building predictive models. Therefore, several strategies can be considered to address this issue. First, extreme values can be clamped to upper bound to reduce the influence of extreme prices. Second, applying a log transformation to *Price* feature can help reduce skewness and make the distribution more suitable for further analysis and modelling. We choose clamping over log transformation because clamping has a direct impact on extreme values while keeping the original scale of *Price* feature. Clamping makes data easy to interpret and improves the robustness of later data modelling.

2.6 NotFullMarketPrice

Feature *NotFullMarketPrice* is viewed as a categorical variable because it has binary values. To be clearer, there is a variety of reasons for a property to be sold not at full market price, such as this transaction only includes a fraction of this property, this transaction is at a reduced price under the Affordable Homes Scheme etc. However, this is not the major scenario of our dataset. Figure 3 shows that nearly 95% of house transactions are at full market price. This helps rule out some edge cases. There are no significant issues identified regarding *NotFullMarketPrice* feature. Since *Price* is the target feature, we can keep other price-related features to provide more information to later research. Thus, we keep *NotFullMarketPrice* feature.

2.7 VATExclusive

Feature *VATExclusive* is treated as a categorical variable because it has binary values. From RPPR, the definition of this feature is that for new properties, their prices are exclusive of a 13.5% VAT. From our dataset, as shown by Figure 4, prices of around 18% of properties do not include the VAT. There are no key issues reported on *VATExclusive* feature. When we assess the target feature, *Price*, it will be thorough for us to consider potential VAT for new properties. Therefore, we propose to keep *VATExclusive* feature for helping reach a robust conclusion.

2.8 DescriptionofProperty

Feature *DescriptionofProperty* is viewed as a categorical variable because the input has finite values. This feature categorized residential properties into three types: New Dwelling house/Apartment, Second-Hand Dwelling house/Apartment, and Teach/Árasán Cónaithe Atháimhe. From Figure 5, around 82% of house transaction is on Second-Hand Dwelling house/Apartment. When we further investigate the *Price* feature, we can provide elaboration on price performance of different types of houses with *DescriptionofProperty*

feature. There are no key issues worth attention. In this way, we propose to keep *DescriptionofProperty* feature.

2.9 PropertySizeDescription

Feature *PropertySizeDescription* is considered as categorical variable because it has finite values of input. This feature categorizes property size. However, the missing rate of nearly 95% is worth discussing. It is clear that we cannot rely on the 5% valid input to drive a robust conclusion. Also, there is no public disclosure of property size for each house transaction. So, it is not easy for us to fill in this feature. Therefore, we propose to drop this feature due to the lack of valid input.

2.10 Others

As Table 2 shows, we identified four duplicate rows of input. For each pair of duplicate rows, they have the same input for every feature. In this way, duplicates may be caused by recording error. We propose to remove all these duplicate rows to prevent double-counting for identical transactions and to ensure the accuracy and integrity of later research.

3. Conclusion

This Data Quality Report evaluates the integrity of the residential property transaction dataset derived from the Residential Property Price Register (RPPR). Overall, the dataset demonstrates a relatively high level of data completeness and consistency across most features, making it suitable for further analysis of the Irish residential property market.

Several data quality issues were identified during our research. These include high cardinality in features such as *DateofSale(dd/mm/yyyy)*, *Address*, and *Eircode*, presence of outliers and extreme values in *Price* feature, a high missing rate in both *PropertySizeDescription* and *Eircode*, and a small number of duplicate records. We need to solve these issues to further interpret the dataset while not bias analytical results.

To address these issues, we discuss several strategies. These include aggregating date information to a monthly level, deriving regional information from address-related features, clamping extreme values in *Price* feature, removing duplicate records, and dropping features with excessive missing values such as *PropertySizeDescription*. These strategies aim to improve data reliability and preserve meaningful information in the dataset.

After applying certain strategies, the dataset is expected to provide a more robust foundation for later analysis and modelling of residential property price patterns in Ireland.

Appendix

A comprehension of data issues

Feature	Type	Data Quality Issue(s)	Impact	Root Cause Analysis	Proposed Solution	Final Action
DateofSale(dd/mm/yyyy)	object	Too much cardinality, not useful for prediction.	High	Natuaral records of dates	Extract year and month	
Address	object	Too much cardinality, not useful for prediction.	High	Natuaral records of different properties	Investigate weather it can be converted to eircode. If not, drop it.	
County	category	No issue	Low	No issue	Keep	
Eircode	object	Too much cardinality, not useful for prediction.	High	Natuaral records of different properties	Try to fill it by address and extract the district from eircode. If can't, drop it.	
Eircode	object	Too much missing(68.86%), not useful for prediction.	High	Data not collected	Try to fill it by address. If can't, drop it.	
Price	float64	Outliers and extremes	Medium	Most high values are valid outliers but several extremes need more investigation	Clamp the price to upper bound	
NotFullMarketPrice	category	No issue	Low	Binary categorical feature	Keep	
VATExclusive	category	No issue	Low	Binary categorical feature	Keep	
DescriptionofProperty	category	No issue	Low	Records of property types	Keep	
PropertySizeDescription	category	Too much missing(97.73%), not useful for prediction.	High	Data not collected	Drop the feature	
Other	none	4 duplicate rows	low	recording error	Remove duplicate rows	

Table 1. An overview of data issues

DateofSale(dd/mm/yyyy)	Address	County	Eircode	Price(€)	NotFullMarketPrice	VATExclusive	DescriptionofProperty	PropertySizeDescription
814	27/09/2016 LISNANAGH, EDGEWORTHSTOWN, LONGFORD	Longford	NaN	140000.0	Yes	No	Second-Hand Dwelling house /Apartment	NaN
2752	27/09/2016 LISNANAGH, EDGEWORTHSTOWN, LONGFORD	Longford	NaN	140000.0	Yes	No	Second-Hand Dwelling house /Apartment	NaN
7404	12/12/2017 26 27 POPES QUAY, NORTH QUAY PLACE, CORK	Cork	NaN	135000.0	No	No	Second-Hand Dwelling house /Apartment	NaN
9273	12/12/2017 26 27 POPES QUAY, NORTH QUAY PLACE, CORK	Cork	NaN	135000.0	No	No	Second-Hand Dwelling house /Apartment	NaN
21308	07/01/2019 THE COACH HOUSE, CIRCULAR RD, DUNMORE EAST	Waterford	NaN	108333.0	Yes	No	Second-Hand Dwelling house /Apartment	NaN
23994	07/01/2019 THE COACH HOUSE, CIRCULAR RD, DUNMORE EAST	Waterford	NaN	108333.0	Yes	No	Second-Hand Dwelling house /Apartment	NaN
48514	12/01/2024 29 KING STREET, BELLTREE, CLONGRIFFIN	Dublin	D13N5XF	570000.0	No	No	Second-Hand Dwelling house /Apartment	NaN
49945	12/01/2024 29 KING STREET, BELLTREE, CLONGRIFFIN	Dublin	D13N5XF	570000.0	No	No	Second-Hand Dwelling house /Apartment	NaN

Table 2. Duplicate data

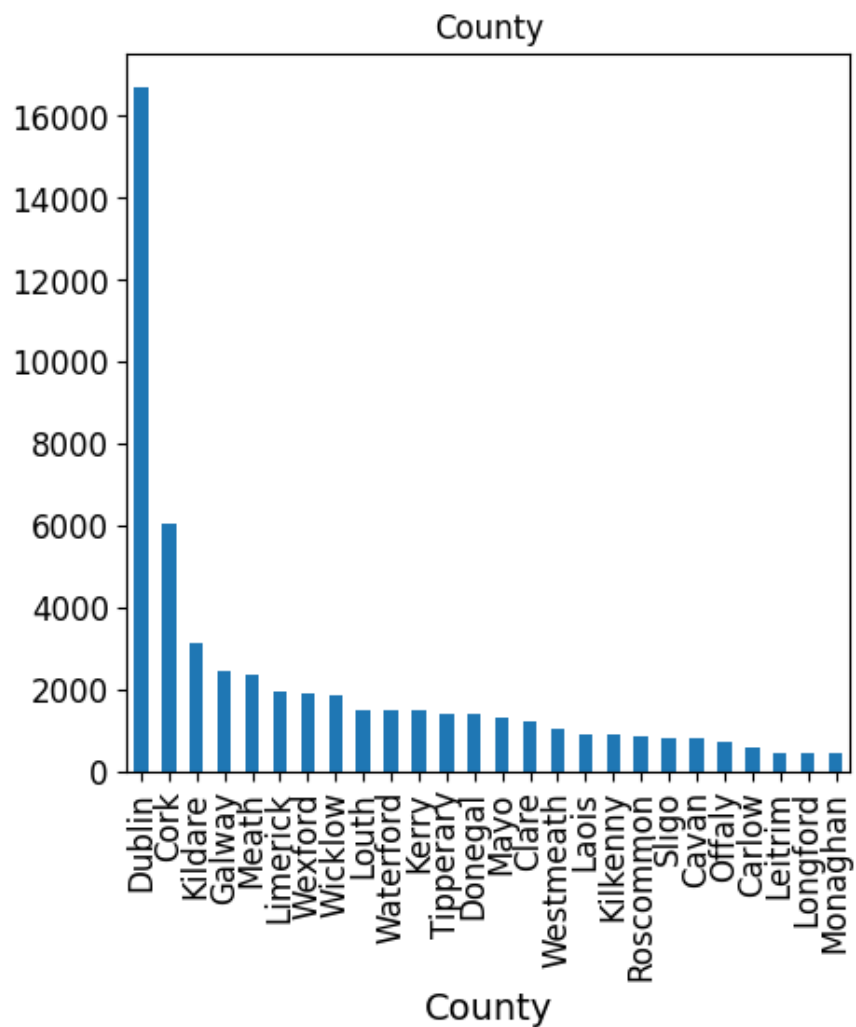


Figure 1. Bar chart on *County*

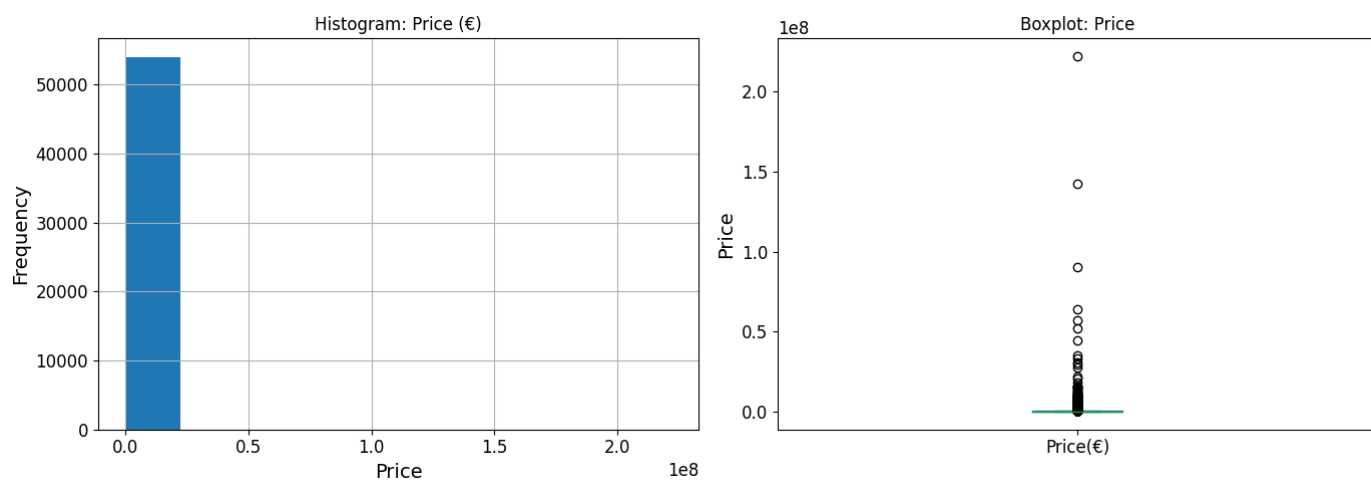


Figure 2. Plot on *Price*

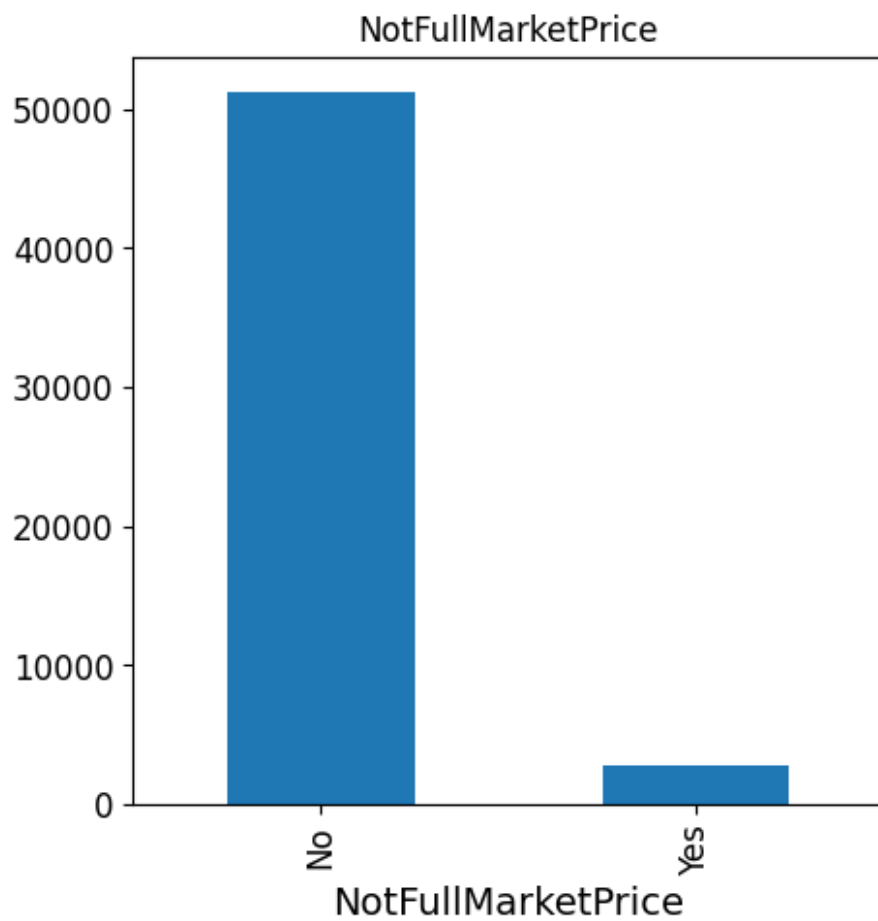


Figure 3. Plot on NotFullMarketPrice

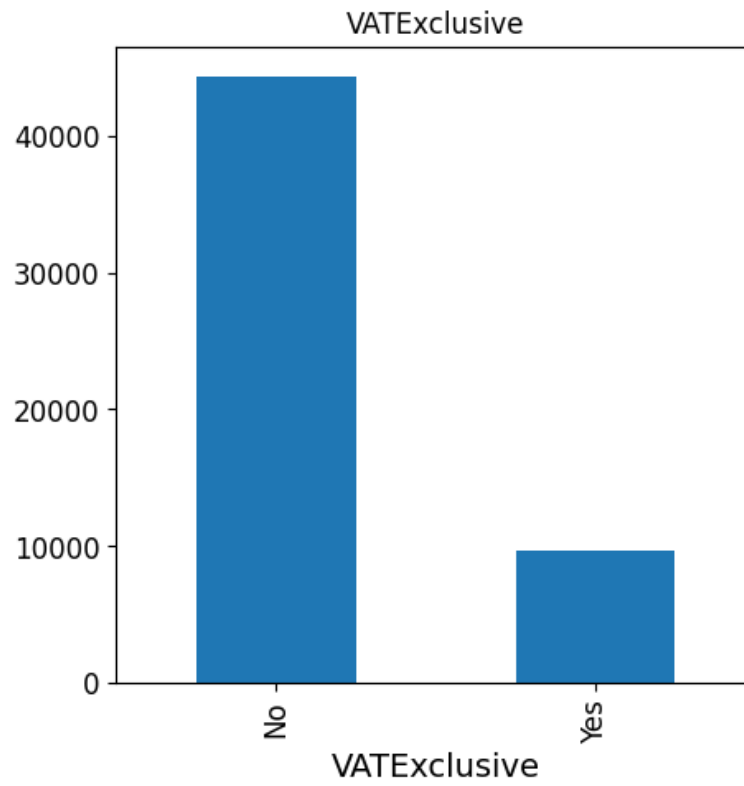


Figure 4. Plot on VATExclusive

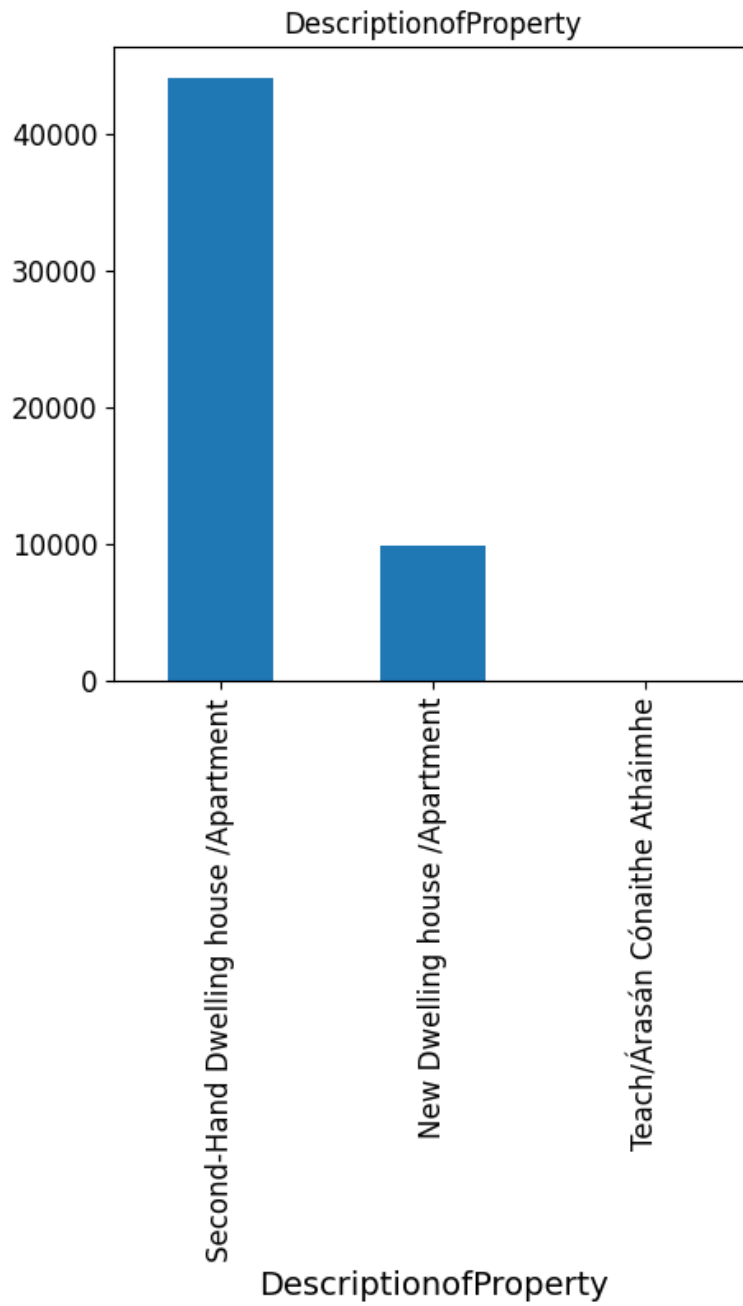


Figure 5. Plot on DescriptionofProperty